

A laboratory that knows what its experiments mean

AICOR Cell 01: vision, research problem, contributions, where we are, and the plan for a first demonstrator

Semantic digital twin of an automated culture-experimentation cell

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Every figure, excerpt and number in this document is generated from the LabAutomationTwin repository (renders from Isaac Sim 6.2, ontologies reasoned by HermiT and Konclude, plans checked by z3). Biology, doses and timings are Cell 01's illustrative surrogate (decision D016); they are not laboratory facts.

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Summary

Robots in biomedical laboratories execute protocols. They do not know what the experiment is for, which culture is the control, which comparison answers the question, or whether a failure at 3 am invalidates a conclusion. Cell 01 is our answer: an automated culture-experimentation cell, built first as a *semantic digital twin*, in which the experiment itself is represented, reasoned about and executed down to the robot motions, with proofs that the motions implement the design.

Where we are. The twin is complete: 5 end-to-end robot episodes run in physics (nominal, identity swap, quarantine and release, imaging, full cell), 110/110 device and robot acceptance tests pass, and AICOR Lab 1 surrounds the cell. Above it sits a stack of knowledge: a lab designator language (LDL) that states what lab actions must achieve, an OWL 2 DL experiment ontology (174 concepts, 246 axioms) whose reasoner classifies a design as adequately controlled or not, a 413-concept lab ontology within the AICOR federation, capability profiles that decide what the lab can run, Bayesian experiment design, and a robot science assistant kernel. A refinement theorem, discharged by machine checks, states that the 111 robot motions of the selected experiment either realize the culture-level design under all constraints or stop in a handler whose guarantee holds.

What is next. A first demonstrator: the challenge–rescue experiment executed end to end in the twin by the generated plans, its cultures imaged by the robot's own camera, the belief updated, faults injected and handled at the level of the experiment, the improved five-arm design carried through the same proof, and one segment on real hardware (Section 7).

1 Vision and mission

Vision. A laboratory in which a robot scientist runs biomedical experiments that are valid by construction: every robot action traces to the scientific purpose it serves, every design is checked for the controls and comparisons its conclusion needs before anything is pipetted, every failure is understood in terms of what it does to the experiment, and every result carries the evidence of how it was obtained.

Mission. To build Cell 01, an automated cell for culture experiments, as the place where this is worked out end to end: a physically faithful, semantically annotated digital twin first, with the knowledge, reasoning and plans that make its robots understand the experiments they perform; then a demonstrator; then real hardware.



Figure 1: AICOR Lab 1 with Cell 01 at its centre (path-traced in Isaac Sim 6.2): operator desks, prep bench, storage, the enclosed cell with its AMR portal.

2 The research problem

How can a robot execute a biomedical experiment so that what it physically does provably implements what the experiment is designed to establish, and so that it knows, at every moment, what its actions, observations and failures mean for the experiment's conclusions?

The problem has three parts:

1. **Meaning across levels.** An experiment is stated over cultures, treatments, controls and comparisons. A robot executes motions over plates, grippers and frames. The robot must carry the experiment's meaning down through these levels and back up: from "the comparator culture receives the challenge at 0 h" to 52 grounded steps in one trip, and from an image of a well back to evidence about a hypothesis.
2. **Validity, not only execution.** A protocol can run perfectly and still answer nothing: no vehicle control, no replicates, a confounded comparison, an unvalidated readout. The robot must judge a design's adequacy before running it, and an execution's validity after.
3. **Living material under uncertainty.** Cultures change while the robot works; they tolerate only limited time outside the incubator; their state is seen only through images that are proxies for viability; contamination and identity swaps happen. Execution must keep the material's constraints and reason about what it does not observe.

3 Why it matters

- **AI scientists now choose experiments, but nothing checks them.** Agent systems propose and run experiments on liquid-handling robots [1]; their checks concern the protocol and the machine (deck state, safe commands), not whether the experiment has the controls and comparisons its conclusion needs.
- **The syntax-to-safety gap.** Commands that parse can be physically unsafe [2]; in biology they can also be scientifically void.
- **Reproducibility.** Much of biomedical research does not reproduce [3]; reporting checklists are applied by people after the fact.
- **Standards stop at devices and data.** SiLA 2, OPC UA LADS and Allotrope make instruments interoperable and data contextual [4]; none represents why a step exists or whether a design can answer its question.
- **Regulated laboratories need traceability.** Every robot action must be verifiable and auditable, as TraceBot set out for sterility testing [5].

4 Why it is hard

Table 1: Why the problem is hard.

difficulty	what it means in Cell 01
The abstraction gap	12 culture actions become 151 grounded steps in 3 trips and 111 robot motions; each level has its own vocabulary, constraints and failure modes.
Time-critical living material	a plate may be outside the incubator for 600 s per excursion (illustrative); culture times (0, 48, 72 h) are fixed by the design.
Partial observability	viability is seen through bright-field images of organoid size, core density and debris; the robot's belief must be calibrated against what it cannot see.
Validity is not execution	a run can be physically perfect and scientifically invalid (an unvalidated imager: realized but not valid evidence).
Failures with consequences	a dropped tip before aspiration is repairable; a misregistered well after the dispense is not; recovery must state what it guarantees.
Devices have inner processes	incubators, liquid handlers and imagers have phases, irreversible points and faults that decide how trustworthy a result is.
Verification across levels	a proof that motions implement a design must compose discrete planning, continuous control, device contracts and biology.
Generality	special devices make automation rigid; general manipulation makes it flexible but harder to verify.

5 Contributions

C1 A semantic digital twin of Cell 01 and AICOR Lab 1 (Isaac Sim 6.2): articulated, physically simulated stations and robots with their own controllers, sensors and monitors, the fit-out, the lab and its site; 5 end-to-end episodes; 110/110 acceptance tests.

C2 A culture digital twin with two separated worlds: hidden simulated biology and the robot's belief about it, connected only by robot records and images; dose ledger, contamination rule, calibrated observation model.

C3 LDL, a lab designator language: actions as contracts (requires, prevents, maintains, achieves, goal) over lab entities only, with instrument roles, device process models, failure handlers with guarantees and an epistemic meta model; a checked abstraction boundary to the realization.

C4 The experiment as a reasoned entity: an OWL 2 DL experiment ontology (174 concepts, 246 axioms) in which HermiT classifies a design by its content: adequately controlled, factorial, interaction-aware; closed-world checks for what OWL cannot express.

C5 A lab ontology in the AICOR federation (413 concepts in 13 modules), bridged to the experiment ontology; the union is consistent and the reasoner proves their arm classes equal.

C6 Lab affordance: capability profiles derived from device records, matched against the requirement classes of designs: what the lab can run, where steps must go, what is missing.

C7 From information gain to motion: Bayesian experiment design selects the experiment; an answer-set program derives the plan; z3 independently validates executability, safety, timing and realization.

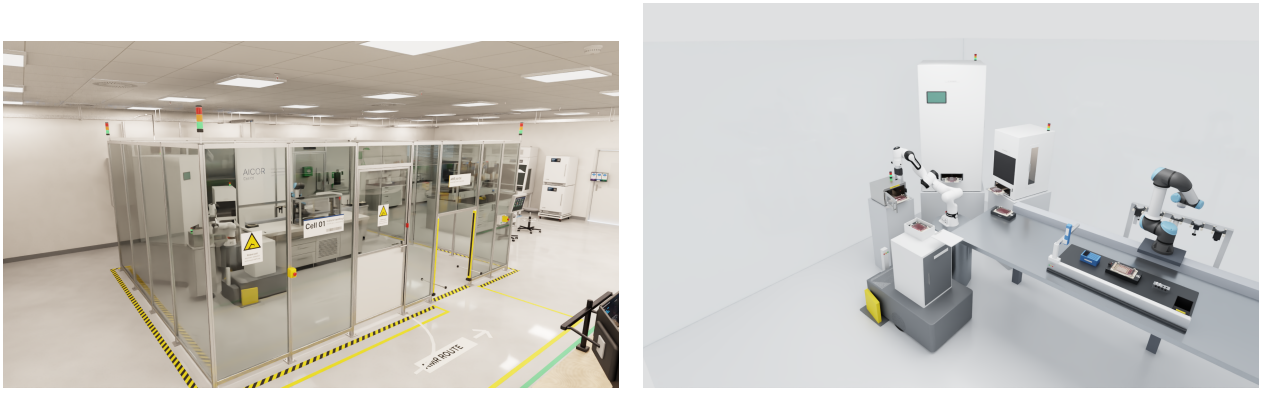
C8 A robot science assistant kernel and CRAM/RPL plans: designators, executive with failure handling, two-column critic (realization vs validity), hash-chained audit trail, competency questions; plans generated as CRAM/RPL and as Giskard motion statecharts.

C9 A refinement theorem with machine-checked obligations: the motions implement the culture-level design or stop in a guaranteed handler, relative to six explicit assumptions; 8 negative controls.

6 Where we are

6.1 The cell and the lab

Cell 01 combines a fixed plate arm (R01, Franka FR3), a rail-mounted flexible arm with an automatic tool changer (R02, UR5e), a clean-sample AMR with a deterministic dock (R03/H2), a plate hotel (L09), a CO_2 incubator (L10), a liquid handler (L12), a bright-field imager (L17), a QC and calibration station (L21) and a quarantine station (H4), inside an interlocked safety enclosure.



the enclosed cell — the stations without the enclosure

Figure 2: Cell 01 with and without its fit-out (path-traced).

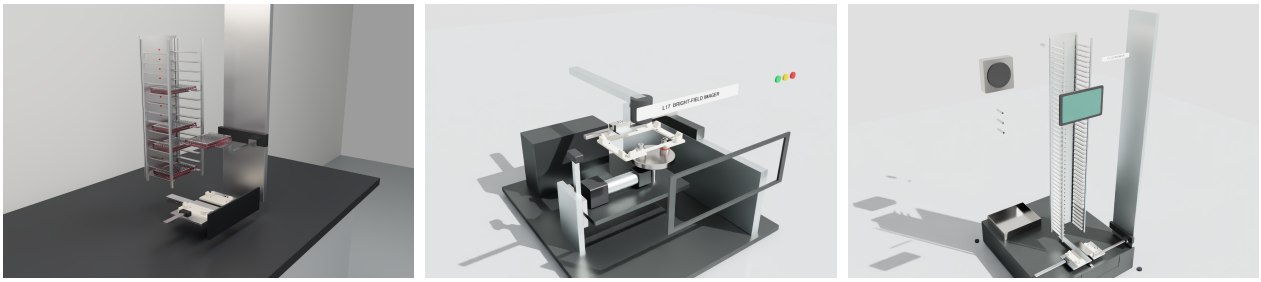
6.2 Devices and their mechanisms

Every station is an articulated model whose controller drives its joints in physics and whose sensors feed its monitors: the L09 hotel's lift, fork and shuttle; the L10 incubator's cassette, lift, gate and environment probes; the L12 gantry; the L17 stage, focus and turret; the H4 door, latch and shuttle.



AMR portal | storage and dock | R02 and its tool magazine

Figure 3: Details of the twin: the AMR route into the cell, storage, and the flexible arm.



L09: fork carrying a plate | L17: stage, focus, illumination | L10: cassette, lift, shuttle

Figure 4: Inside the devices (housings removed): the mechanisms the controllers drive.

6.3 Robots in action

The episodes run the cell's robots and devices in PhysX at 240 Hz with the controllers in the loop; their run logs become Cell 01 NEEM records through Cell 01's own builders and validator.

6.4 The cultures

The culture twin keeps two worlds apart. The *simulated world* holds the hidden biology of every well (injury, protective response, reserve, viability; Cell 01's surrogate). The *belief* is a particle filter per well, fed only by robot records (committed doses) and 4× images. Its 90 % intervals contain the hidden viability in every snapshot of every run.

The robot's own images. The L17 controller images the simulated cultures through its optics model: stage motion, objective change, autofocus, exposure and monitors as in an episode. Without plate registration the field lands 0.2–0.65 mm off the well centre and 7 of 20 frames are valid. Edge-finding plate registration measures the walls of two wells consistently (shift (-0.03, -0.25) mm, rotation +0.10°), but it does not yet correct the field: 7 of 20 frames are valid and the field still lies 0.32–0.58 mm from the well centre. The offsets are stable over time, so the plate does not slide; the mismatch lies between where the controller believes the field is and where the optics images, and is under investigation (WP2).

6.5 Knowledge: ontologies and the designator language

LDL: the contract of an intervention (rsa/registry/lab/contracts.ldl)



L12 gantry and deck | R02 tool magazine | H4 quarantine

Figure 5: Liquid handling, tools and containment.



identity fault at L10 carrying to quarantine quarantined and locked

Figure 6: The identity-swap episode: the wrong plate is detected at the incubator and contained in H4.



Figure 7: The improved five-arm design, one culture per arm at 0, 24, 48 and 72 h (path-traced camera view of the simulated cultures).

Table 2: End-to-end episodes in the twin (test_reports/episodes).

episode	route	duration	outcome
cell01-ep-001-nominal	store	66.3 s	stored: True
cell01-ep-002-identity_swap	quarantine	49.7 s	contained: True, reentry_blocked: True
cell01-ep-003-quarantine_release	release_to_L10	62.6 s	incubated: True
cell01-ep-004-imaging	imaging	243.2 s	images_valid: 3, stored: True
cell01-ep-005-full_cell	full_cell	743.5 s	qc: ['out_of_tolerance', 'pass', 'pass'], images_valid: 2, incubated: True

Table 3: Device and robot acceptance tests (physics in the loop): 110/110 pass; H4 robot loading: 8/8 seated.

acceptance test	passed
h4 physics	14/14
l09 physics	11/11
l10 physics	11/11
l12 physics	10/10
l12 robot	7/7
l17 physics	15/15
l17 robot	6/6
l21	10/10
r02	10/10
r03	9/9
site route	7/7

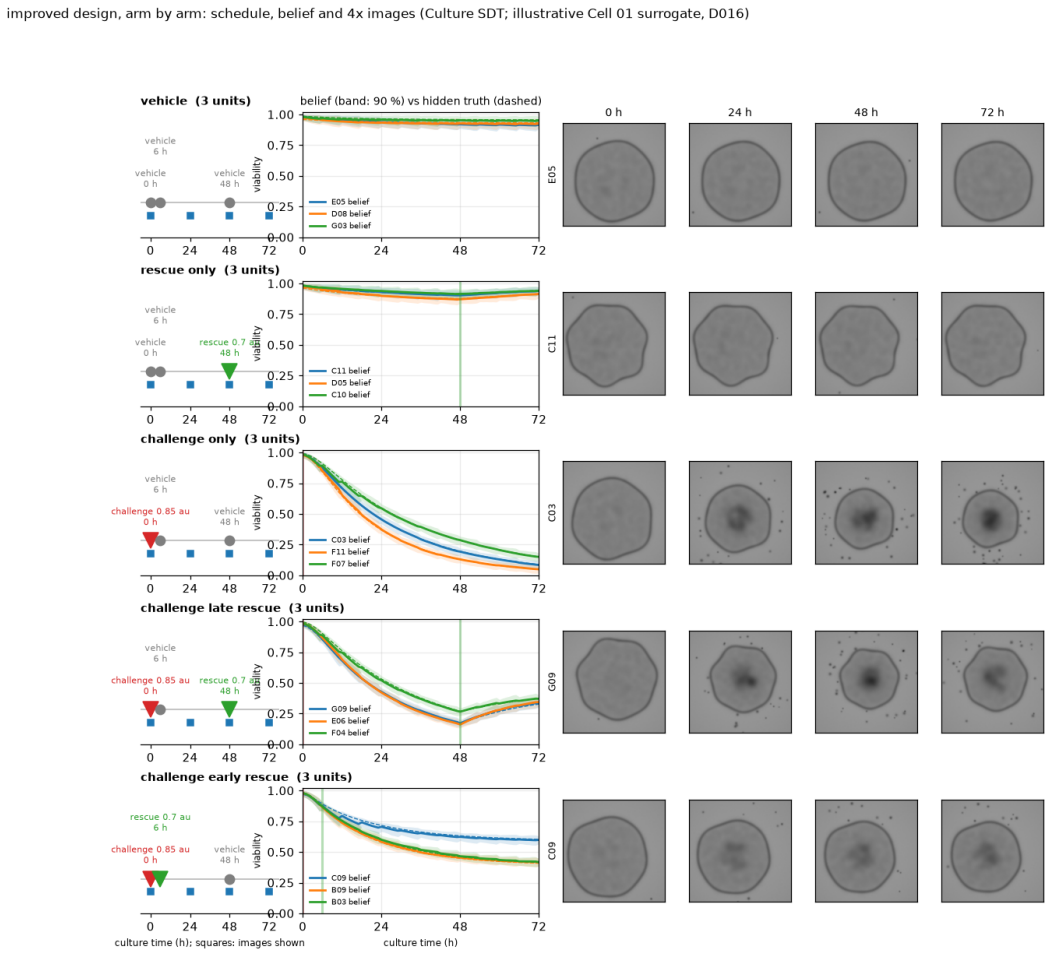


Figure 8: Arm by arm: what is given when, the twin's belief about each culture's viability against the hidden truth, and the 4× images.

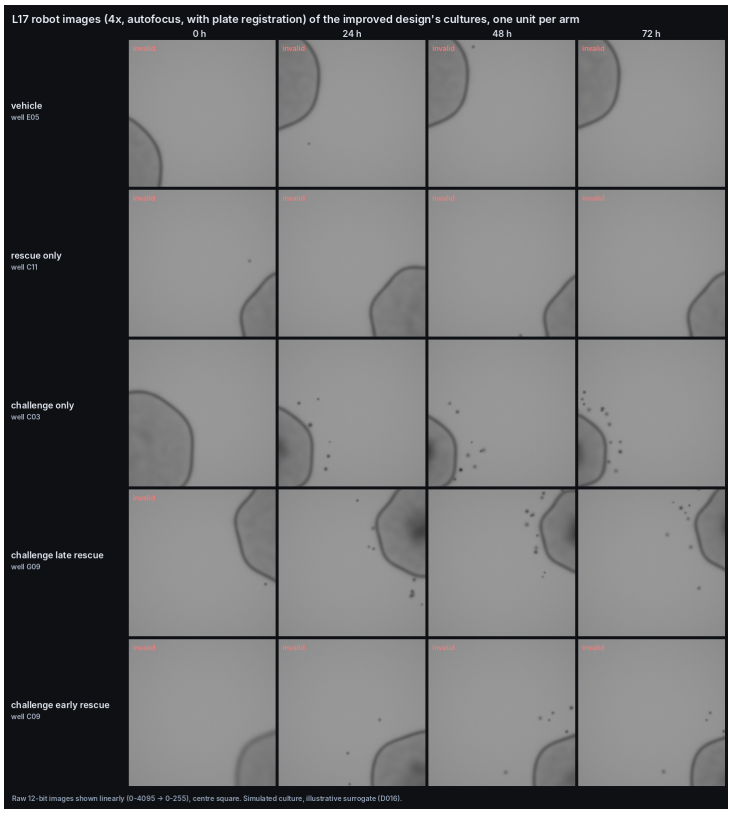


Figure 9: The robot's own L17 images of the improved design's cultures (4×, autofocus).

```
(contract intervening
  (roles (into (the well ?w (of ?p))) (substance ?s) (dose ?q) (at (culture-time ?t)))
  (instrument (an instrument ?d
    (slot (a plate-slot (accepts ?p)))
    (process (a process (type substance-transfer) (into ?w) (of ?s) (amount ?q)))))
  (goal (the well ?w (received ?s ?q)) :tolerance 5 % :column validity)
  (requires
    (known (the plate ?p (identity verified))) :consequential
    (known (some substance ?s (identity verified))) :consequential
    (believed (the plate ?p (contamination none)) (grade (source inferred))) :consequential
    (the plate ?p (culture-time ?t)))
  (prevents
    (an event (type contamination-introduced) (of ?p)))
  (scope (begins-when (a portion ?u (of ?s) (drawn-for ?self)))
         (ends-when (the portion ?u (delivered true)))
    (prevents (a transfer (of ?u) (into (a well (of ?p) (not ?w))))))
  (achieves
    (a transfer (of ?s) (into ?w))
    (a record (of ?self) (states ?w ?s ?q ?t) (original true)))
  (signals
    (a failure (type identity-mismatch))
    (a failure (type dose-not-delivered))
```

LDL: the improved design is authored once and read by every layer (experiment_improved.ldl)

```
  (factor (a factor (name challenge)
    (level (a level (name vehicle)
      (administration (an administration (agent vehicle) (dose 0 au) (at (culture-time 0 h))))))
    (level (a level (name c085)
      (administration (an administration (agent challenge) (dose 0.85 au) (at (culture-time 0 h)))))))
  (factor (a factor (name rescue_time)
    (level (a level (name none)
      (administration (an administration (agent vehicle) (dose 0 au) (at (culture-time 6 h))))
      (administration (an administration (agent vehicle) (dose 0 au) (at (culture-time 48 h))))))
    (level (a level (name h6)
      (administration (an administration (agent rescue) (dose 0.70 au) (at (culture-time 6 h))))
      (administration (an administration (agent vehicle) (dose 0 au) (at (culture-time 48 h))))))
    (level (a level (name h48)
      (administration (an administration (agent vehicle) (dose 0 au) (at (culture-time 6 h))))
      (administration (an administration (agent rescue) (dose 0.70 au) (at (culture-time 48 h)))))))
```

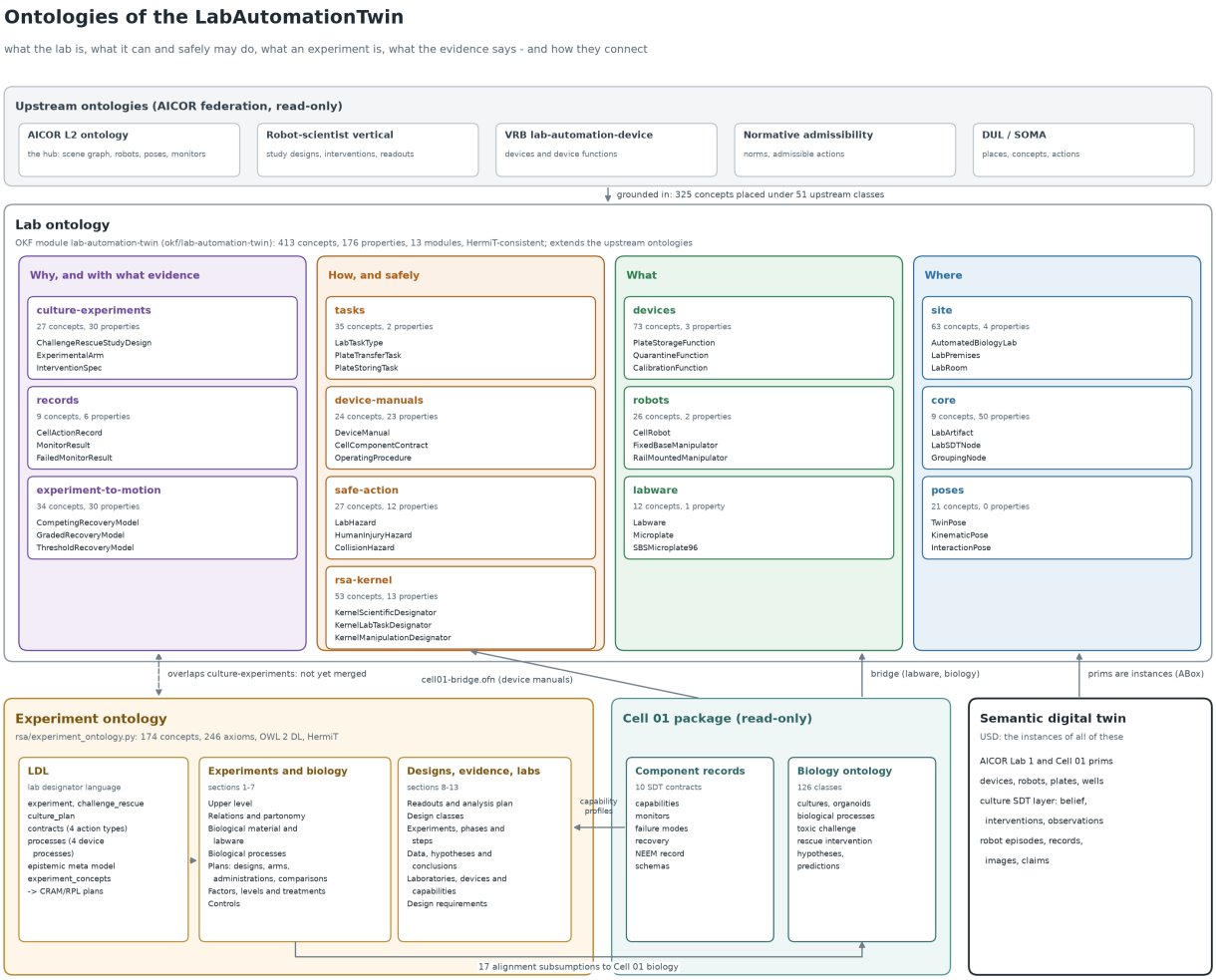


Figure 10: The ontologies and how they connect: upstream AICOR federation, the lab ontology's 13 modules, the experiment ontology and LDL, Cell 01's records and biology, and the twin as their instances.

```
(arm (an arm (name vehicle) (units 3) (at-level (challenge vehicle) (rescue_time none))))
(arm (an arm (name rescue_only) (units 3) (at-level (challenge vehicle) (rescue_time h48))))
```

The experiment ontology. A design is classified from its content. Selected definitions (OWL 2 DL, the reasoner decides membership):

$$\mathsf{NegativeControlArm} \equiv \mathsf{UntreatedArm} \sqcup \mathsf{VehicleControlArm}$$

$$\mathsf{ChallengeOnlyArm} \equiv \mathsf{ArmSpecification} \sqcap \exists \mathit{hasAdministrationSpec}.\mathsf{ChallengeSpec} \sqcap \neg\exists \mathit{hasAdministrationSpec}.\mathsf{RescueSpec}$$

$$\mathsf{RescueEffectComparison} \equiv \mathsf{ComparisonSpecification} \sqcap \exists \mathit{compares}.\mathsf{ChallengeRescueArm} \sqcap \exists \mathit{compares}.\mathsf{ChallengeOnlyArm}$$

$$\mathsf{AdequatelyControlledInjuryRescueDesign} \equiv \mathsf{AgentControlledInjuryRescueDesign} \sqcap \mathsf{PositiveControlledDesign} \sqcap \mathsf{ReplicatedDesign} \sqcap \mathsf{WellContrastedDesign} \sqcap \mathsf{PreSpecifiedDesign}$$

$$\mathsf{TreatmentAndImagingModule} \equiv \mathsf{DosingLab} \sqcap \mathsf{CultivationLab} \sqcap \mathsf{BrightfieldImagingLab} \sqcap \mathsf{ValidatedSizeReadoutLab}$$

$$\mathsf{RequiresDosing} \equiv \mathsf{StudyDesign} \sqcap \exists \mathit{hasArm}.\exists \mathit{hasAdministrationSpec}.\mathsf{AdministrationSpecification}$$

HermiT classifies the current design as a validated injury–rescue design but *not* adequately controlled (no rescue-only or vehicle arm, no positive control, no replicates); the improved design is adequately controlled, interaction-aware and has a therapeutic-window comparison. Each variant that removes one ingredient loses exactly the corresponding class.

One experiment model. The lab ontology's design concepts are bridged to the experiment ontology. Konclude finds the union consistent with 0 unsatisfiable classes and infers lab:ChallengeOnlyArm ≡ xp:ChallengeOnlyArm; lab:ChallengeRescueArm ≡ xp:ChallengeRescueArm; lab:ControlArm ≡ xp:UntreatedArm.

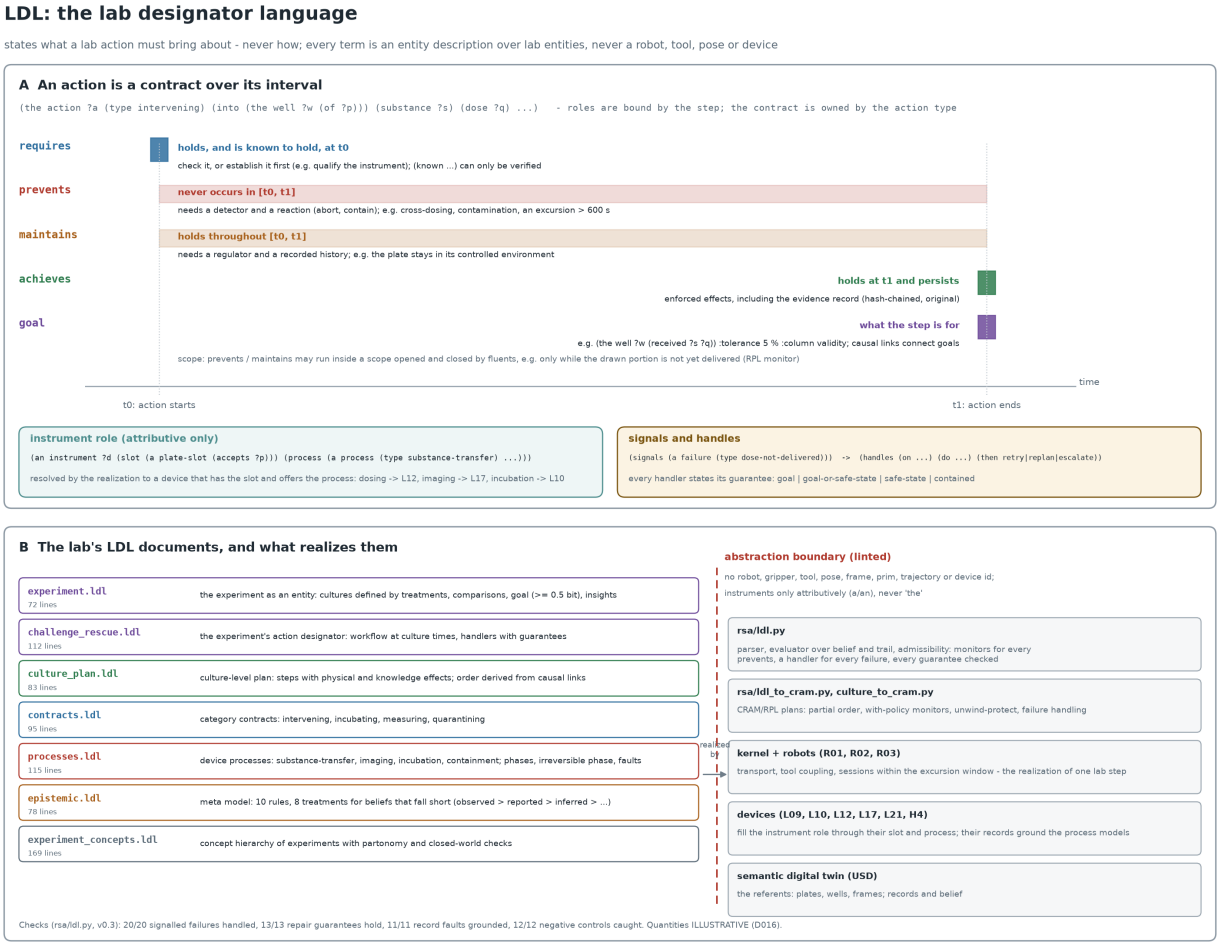


Figure 11: LDL: an action as a contract over its interval, and the lab's LDL documents above a checked abstraction boundary.

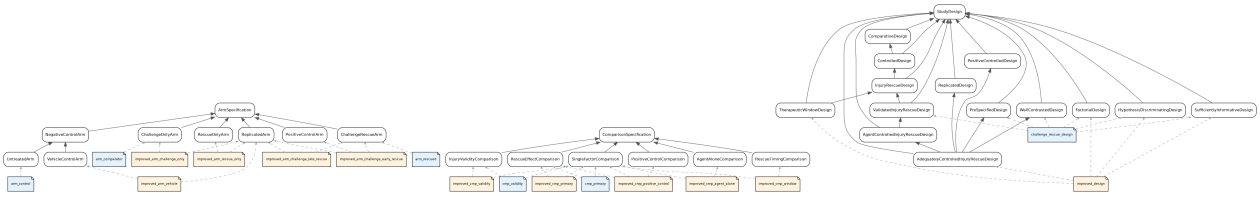


Figure 12: The inferred taxonomy (HermiT) with both designs realized under it.

6.6 From required information to robot motion

Three hypotheses compete: graded recovery, a threshold, and a protective response. The design is chosen for the expected information gain it yields about them (0.544 bit, at least 0.5 bit required); an answer-set program derives the plan (sessions, trips, steps); z3 checks it independently.

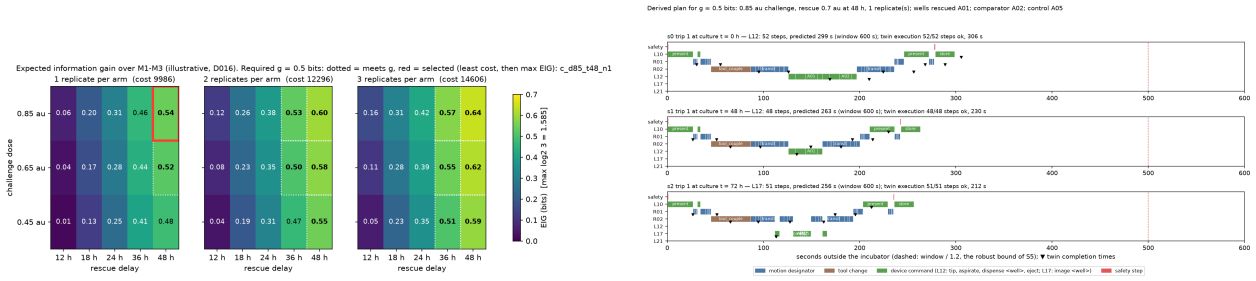
The culture-level plan as a CRAM/RPL program (generated; rsa/out/culture_plan_src.lisp, first lines)

```
;;; -*- Mode: Lisp -*-
;;; Challenge-rescue experiment at the culture level: the top-level CRAM/RPL plan.
;;; Generated by rsa/culture_to_cram.py from rsa/registry/lab/culture_plan.ldl (LDL v0.3, culture level):
;;; 12 culture action designators, 13 causal links (7 physical, 6 knowledge),
;;; 13 order constraints, 6 action types. Below it: the lab level (rsa/out/challenge_rescue_src.lisp),
;;; whose action plans perform the lab action designators the culture actions are refined into.
;;; No plate, well, instrument or robot appears in code: vessels come from the mapping at run time.
;;; Notation: RPL control structures in CRAM Lisp syntax. Not executed: no Lisp CRAM here. ILLUSTRATIVE (D016).

(in-package :culture-src)   ; uses :lab-src

;;; The culture level adds this vocabulary to RPL/CRAM and to the lab level (package :lab-src):
;;; HOLDS, PREVENT, MAINTAIN, ENSURE, DEF-ACTION-PLAN and the lab action plans come from there.

(defmacro with-culture-contract ((self) &body refinement)
```



expected information gain of the candidates

the derived plan: sessions and trips

Figure 13: Experiment to motion: choosing the experiment by information gain, and the plan derived for it.

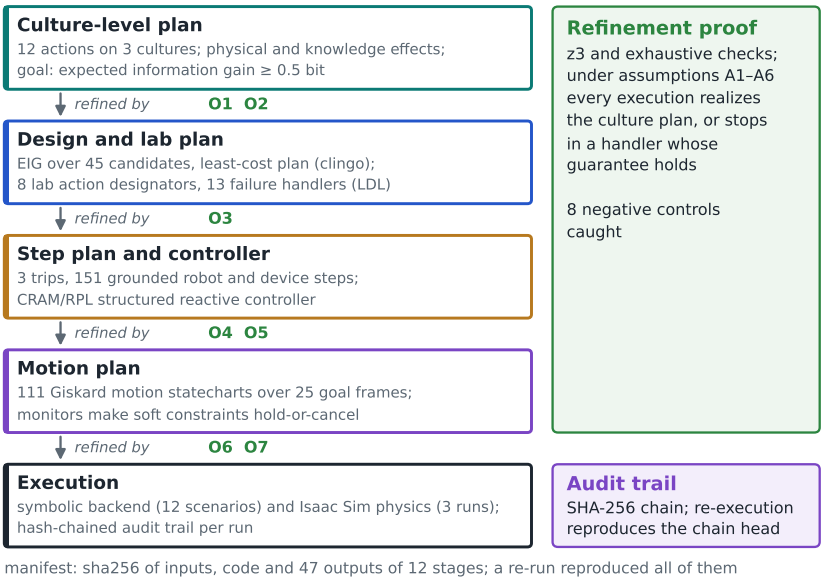


Figure 14: The chain from the experiment to the robots, as the kernel executes it.

```
  "The clauses a culture action designator carries: wait for each requires-item a causal link
supplies, achieve the :establishable ones, perform the REFINEMENT, then add the PHYSICAL effects
to the belief state as projected (graded by the culture model until an observation bears on
them) and assert the KNOWLEDGE effects (records, images, estimates, the belief update) as known."
  `(seq (dolist (r (desig-prop-values ,self :requires))
          (if (establishable-p r) (achieve r) (wait-for (holds r))))
        ,@refinement
        (project-effects (desig-prop-values ,self :physical) :grade '(source projected))
        (assert-effects (desig-prop-values ,self :knowledge) :grade '(source recorded))))

(defmacro ensure-given (description)
  "A given-item: verified before the culture plan starts; a shortfall is treated by the
epistemic meta model like any ENSURE."
  `(ensure ,description :grade (source verified) :consequential t))
```

6.7 Guarantees: the motions implement the design

Theorem 1 (Refinement). *Under assumptions A1–A6 (motion-framework semantics, monitor soundness, mechanical capture, grasp, device contracts, reachable goal frames), every execution of the 111 robot motions of the selected plan either (i) realizes the culture-level design, every treatment and observation to the right culture at the right culture time in the designed order, the control never exposed, and every stated constraint kept; or (ii) is stopped by a monitor and ends in a failure handler whose stated guarantee holds.*

6.8 What the lab affords

Capability classes of a laboratory are defined by the functions its devices bear; requirement classes of a design by its content. The reasoner classifies both, and a closed-world matcher decides what can run where.

Table 4: The seven obligations; each is discharged by a machine check, and 8 negative controls show the checks discriminate. Full statements and proofs: report/proofs/refinement_theorems.pdf.

	obligation	status
O1	culture plan well formed	holds
O2	abstraction: motion-level effects entail the culture plan	holds
O3	lab plan: order, threats, instruments, handlers, guarantees	holds
O4	step level: executability, safety, time, realization (z3 V1-V5)	holds
O5	step level: cleanliness	holds
O6	motion refinement: 111 Giskard motions	holds
O7	failure branch: every cancellation reaches a guaranteed handler	holds

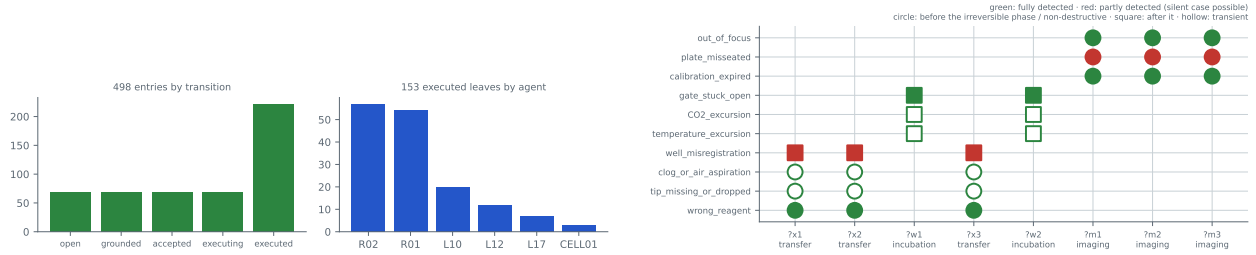


Figure 15: Evidence and trust: every lifecycle transition is a hash-chained record; device processes decide how far a result can be trusted.

Table 5: The kernel on twelve scenarios (symbolic backend): repairs, escalations, refusals before execution, and realized-but-invalid evidence (imager swapped).

scenario	what is changed	end	realization / validity	as expected
nominal	the derived 0.5-bit experiment, fault-free	executed	ok / ok	yes
uncalibrated known	the KB does not assert calibrated(l17); a QC calibration is planned before imaging (step insertion by knowledge)	executed	ok / ok	yes
calibration lapsed	the KB believes L17 calibrated but it lapsed; detected at imaging, repaired by QC and re-imaging	executed	ok / ok	yes
tip lost	the tip falls off after the pick at the rescue dose; detected before aspiration, tip replaced	executed	ok / ok	yes
identity swap	L10 holds P18 instead of P17 at the rescue session; quarantine P18, escalate	escalated	x / x	yes
tool changer slow	the tool changer takes 600 s at the rescue session; window exceeded, safe return	escalated	x / x	yes
imager swapped	L17 replaced by a capability-matched imager without a validated binding for the readout (device exchange)	executed	ok / x	yes
imager incapable	L17 replaced by a device without well imaging; the measurement session cannot be grounded	abstained	x / x	yes
rescue unbound	the rescue reagent is not bound to a rack position; the rescue dose cannot be grounded	abstained	x / x	yes
control dropped	the workflow is changed to drop the control well; the experiment is refused before any motion	abstained	x / x	yes
contamination	contamination is suspected; challenge and rescue are refused	abstained	x / x	yes
card incomplete	the dose category card lacks its evidence obligations; the understanding gate refuses the dose	abstained	x / x	yes

Table 6: Design verdicts against the site.

design	verdict	missing
the current challenge-rescue design	executable in Cell 01	—
the improved design (vehicle, rescue-only, early rescue, 3 replicates)	not executable as modelled	reagent not stocked in cell01: vehicle
the improved design with an ATP luminescence endpoint at 72 h	not executable as modelled	reagent not stocked in cell01: vehicle
a challenge x oxygen design (environment factor)	not executable as modelled	RequiresMultipleEnvironments: no laboratory is a MultiEnvironmentLab

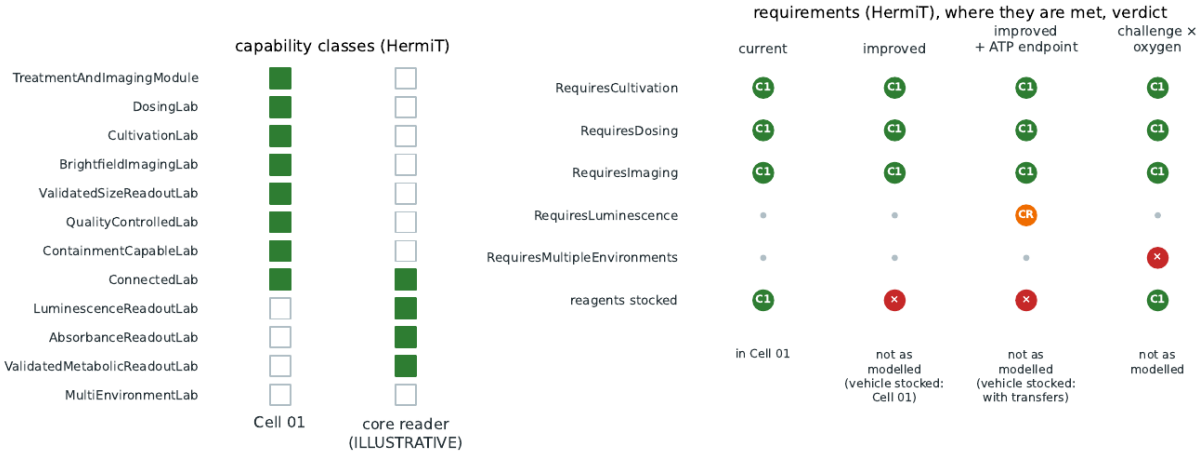


Figure 16: Capability classes of Cell 01 and an (illustrative) core plate reader; the requirements of four designs and where they are met.

6.9 Status at a glance

Table 7: Where the work stands.

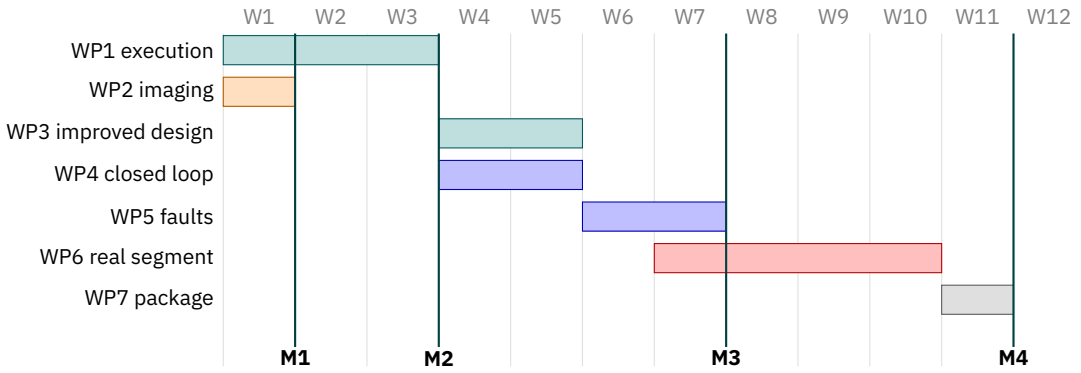
component	status	evidence
Twin of Cell 01 and Lab 1	built, tested	110/110 tests; 5 episodes
Culture twin (belief vs truth)	built, audited	90 % coverage 1.00 in all runs
Robot imaging of cultures	built; registration not yet effective	7/20 frames valid; field 0.3-0.6 mm off
LDL	built, checked	20/20 failures handled, 13/13 guarantees, 12/12 negative controls
Experiment ontology	built, reasoned	174 concepts, 246 axioms; 8/8 negative controls
Lab ontology + bridge	built, reasoned	413 concepts; union consistent
Lab affordance	built	4 designs judged; vehicle stock missing
Experiment to motion	built, z3-validated	7 plans derived, 3 refused
Kernel	runs (symbolic, Isaac)	12/12 scenarios as expected
CRAM/RPL and Giskard plans	generated, checked; not executed	no CRAM runtime installed
Refinement theorem	proved (relative to A1-A6)	current design only
Real hardware	not started	—

7 Plan for a first demonstrator

Goal. Demonstrator D1 shows, on Cell 01's twin and in one real segment, that a robot executes a biomedical experiment it understands: the challenge–rescue experiment is designed, checked for adequacy, matched to the lab, planned, executed by the generated plans, observed with the robot's own camera, interpreted into an updated belief, and reported with its evidence, and its failures are handled at the level of the experiment, with the refinement theorem holding for the executed design.

Table 8: Work packages (durations proposed, to be agreed).

WP	name	content	weeks
WP1	Execution	Run the generated plans in the twin: a CRAM (or PyCRAM) executive or the kernel's Isaac backend executing the designators and Giskard motions; complete the full-cell run.	3
WP2	Valid imaging	Find the field-position mismatch that registration does not remove (stage-to-optics calibration); robot images of every arm valid; update assumption A5.	1-2
WP3	Improved design, full chain	Derive the step and motion plans for the five-arm design (15 cultures, split dosing sessions); extend the refinement proof to it.	2
WP4	Closed loop	The culture twin fed by the robot's own images; belief update; the next experiment chosen by information gain; report and provenance generated.	2
WP5	Faults at experiment level	Injected faults (identity swap, lost tip, calibration lapse, contamination, lost culture) mapped to the comparisons and conclusions they affect; partial reruns.	2
WP6	One real segment	One step on real hardware (plate transfer or imaging) predicted by the twin and compared.	2-4
WP7	Package	Reproducibility pipeline with all generators; merge to main; demonstrator video and report.	1



M1 (week 1)
the robot's images of every arm valid with registration.

M2 (week 3)
the current design executed end to end in the twin by the generated plans; O2 checked on the executed trail, not only on the plan.

M3 (week 7)
the improved design carried through the proof; fault campaign with experiment-level consequences; the closed loop proposes the next experiment.

M4 (week 11)
one real segment against the twin's prediction; demonstrator package.

Table 9: Acceptance criteria of D1.

criterion	measure
Faithful execution	the executed trail satisfies every culture-level obligation (O2 on the trail) in every nominal run
Valid observation	share of robot images valid; field offset from the culture within the 4× field
Proof coverage	refinement theorem regenerated and holding for both designs
Robustness	every injected fault ends in a handler with its guarantee; the conclusions it invalidates are named
Closed loop	belief updated from the robot's images; next experiment proposed with its expected gain
Twin vs real	the real segment's duration and outcome against the twin's prediction (target set after the first measurement)

Table 10: Risks.

risk	mitigation
No CRAM runtime on this machine	use PyCRAM or the kernel's Isaac backend as the executive for D1; keep the CRAM/RPL output as the reference
One 8 GB GPU: concurrent Isaac jobs crash it	serialize GPU jobs in the build pipeline
Integration of Giskard with the twin	start with R01 legs; fall back to the twin's own planners for R02
Real hardware availability	decide the platform early (week 2); the twin demonstrator stands on its own
Biology realism	keep the surrogate labelled; invite scientists to pose the question for a later demonstrator

Decisions needed. (1) the executive for D1: Lisp CRAM, PyCRAM, or the kernel's Isaac backend; (2) the real-hardware segment and platform; (3) the audience and date of the first showing.

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